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9. Trace

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 Calculator

Definition 9.1 The **trace** of a square matrix **A** is the sum of the entries along the main diagonal. That is, for

$$\mathbf{A} = \begin{pmatrix} a_{11} & \cdots & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & & \ddots & \vdots \\ a_{n1} & \cdots & \cdots & a_{nn} \end{pmatrix}.$$

the trace is

$$\text{tr} \mathbf{A} = a_{11} + a_{22} + \cdots + a_{nn}.$$

Example 9.2 If $\mathbf{A} = \begin{pmatrix} 4 & 6 & 9 \\ 1 & 7 & 8 \\ 2 & 3 & 5 \end{pmatrix}$, then $\text{tr} \mathbf{A} = 4 + 7 + 5 = 16$.

Warning: Recall that trace and determinant make sense only for **square** matrices.

Problem 9.3 For an $n \times n$ matrix **A**, how do $\text{tr}(-\mathbf{A})$ and $\det(-\mathbf{A})$ relate to $\text{tr} \mathbf{A}$ and $\det \mathbf{A}$?

Solution

Negating **A** negates in particular all diagonal entries of **A**, so $\text{tr}(-\mathbf{A}) = -\text{tr} \mathbf{A}$.

On the other hand, negating **A** amounts to multiplying every row by -1 , which multiplies $\det \mathbf{A}$ by $(-1)^n$ because there is one factor of -1 for each row. Thus

- If n is even, then $\det(-\mathbf{A}) = \det \mathbf{A}$.
- If n is odd, then $\det(-\mathbf{A}) = -\det \mathbf{A}$.

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Trace of sums

1/1 point (graded)
If **A** and **B** are both 5×5 matrices and $\text{tr} \mathbf{A} = 7$, $\text{tr} \mathbf{B} = 16$, what is the value of $\text{tr}(\mathbf{A} + \mathbf{B})$?

$\text{tr}(\mathbf{A} + \mathbf{B}) =$ ✓ Answer: 23

Solution:

Denote the entries of **A** and **B** at the i^{th} row and j^{th} column by a_{ij} and b_{ij} respectively. Then

$$\begin{aligned} \text{tr}(\mathbf{A} + \mathbf{B}) &= (a_{11} + b_{11}) + (a_{22} + b_{22}) + \cdots + (a_{55} + b_{55}) \\ &= (a_{11} + \cdots + a_{55}) + (b_{11} + \cdots + b_{55}) \\ &= \text{tr} \mathbf{A} + \text{tr} \mathbf{B} = 7 + 16 = 23. \end{aligned}$$

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ⓘ Answers are displayed within the problem

Trace of scalar multiple

1/1 point (graded)
Let **A** be an 5×5 matrix. If $\text{tr} \mathbf{A} = -2$, what is the trace of $4\mathbf{A}$?

$\text{tr}(4\mathbf{A}) =$ ✓ Answer: -8

Solution:

Again denote the entries of $\mathbf{A}$ at the i^{th} row and j^{th} column by a_{ij} . Then

$$\text{tr}(4\mathbf{A}) = 4a_{11} + \cdots + 4a_{55} = 4(a_{11} + \cdots + a_{55}) = 4\text{tr}(\mathbf{A}) = -8.$$

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Trace of identity matrix

1/1 point (graded)
Let $\mathbf{I}$ be the $n \times n$ identity matrix. Compute $\text{tr}(\mathbf{I})$.

tr(I) =

 Answer: n

Solution:

$$\text{tr}(\mathbf{I}) = \underbrace{1 + \cdots + 1}_{n \text{ times}} = n.$$

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Trace and determinant of products of matrices

2/2 points (graded)
Let $\mathbf{A}, \mathbf{B}$ be $n \times n$ matrices.

True or False? The trace of $\mathbf{AB}$ is the product of the individual traces:

$$\text{tr}(\mathbf{AB}) = (\text{tr}\mathbf{A})(\text{tr}\mathbf{B})$$

☐ True

☒ False



True or False? The determinant of $\mathbf{AB}$ is the product of the individual determinants:

$$\det(\mathbf{AB}) = (\det\mathbf{A})(\det\mathbf{B}).$$

☒ True

☐ False



Solution:

- The trace of $\mathbf{AB}$ is **NOT** the product of the individual traces. Consider the matrix $\mathbf{J} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$. We see that $\mathbf{J}^2 = -\mathbf{I}$, so $\text{tr}(\mathbf{J}^2) = -2$ even though $\text{tr}(\mathbf{J}) = 0$.
- We told you in the previous lecture that $\det(\mathbf{AB}) = (\det\mathbf{A})(\det\mathbf{B})$.

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